NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

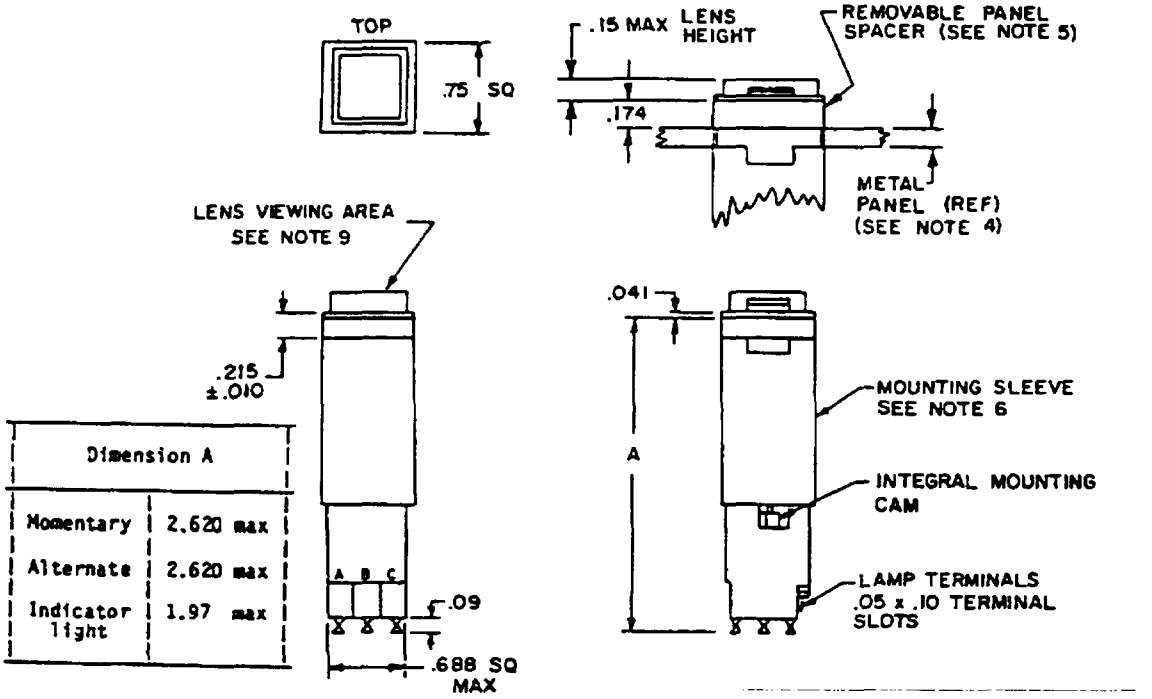
INCH-POUND
MIL-PRF-22885/67F
18 December 1987
SUPERSEDING
MIL-S-22885/67E
13 November 1978

PERFORMANCE SPECIFICATION SHEET

SWITCHES, PUSH BUTTON, ILLUMINATED, 4-LAMP, 0.75 SQUARE, 7 AMPERES, AND LOW LEVEL ELECTROMAGNETIC INTERFERENCE SHIELDED, SUNLIGHT READABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the switches described herein shall consist of this specification and the latest issue of MIL-S-22885.



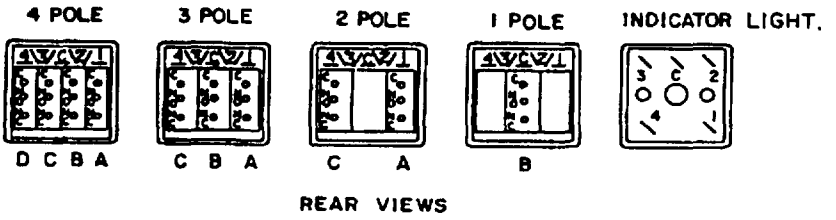
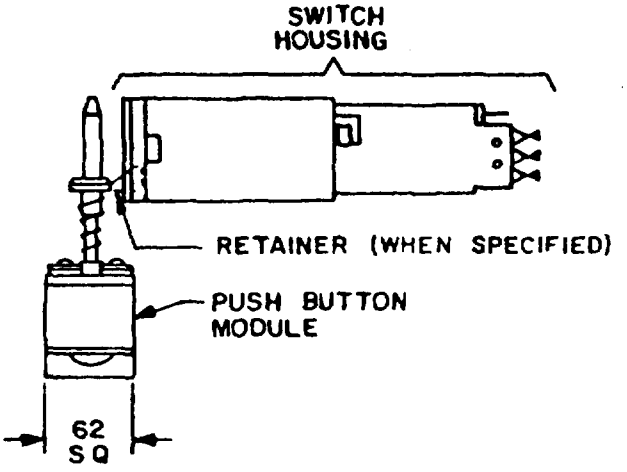
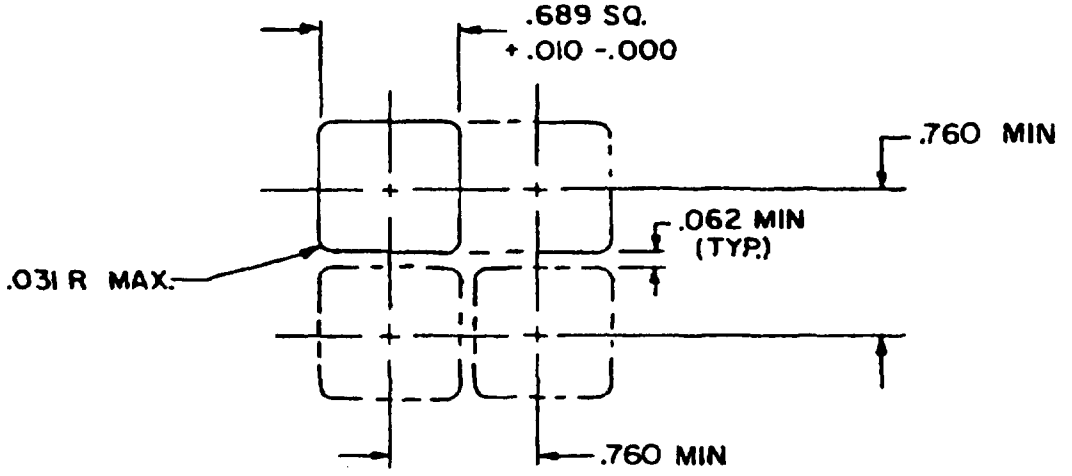


FIGURE 1. Design and construction.

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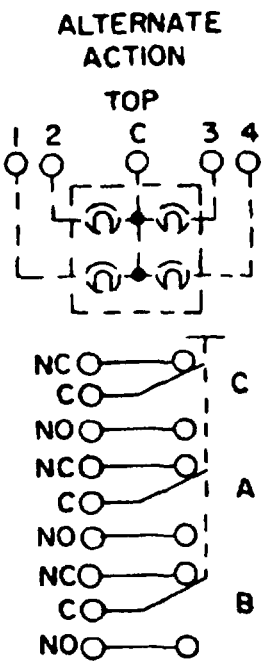


RECOMMENDED PANEL CUTOUT
FOR SINGLE UNIT WHEN MATRIX MOUNTING

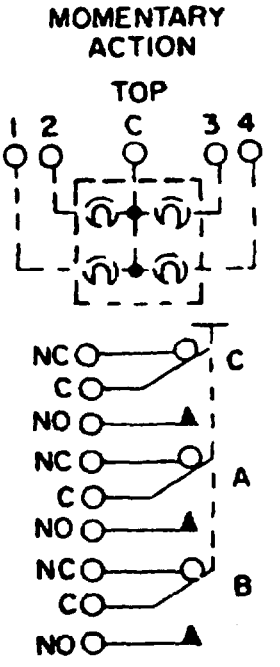
Inches	mm	Inches	mm
.010	0.25	0.53	13.5
.030	0.76	0.62	15.7
.031	0.79	0.688	17.48
.041	1.04	0.689	17.50
.05	1.3	0.75	19.0
.062	1.57	0.760	19.30
.09	2.3	1.97	50.0
.15	3.8	2.024	51.41
.174	4.42	2.620	66.55
.215	5.46		

FIGURE 1. Design and construction - Continued.

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SWITCH POSITIONS 3PDT
(SHOWN FOR REFERENCE)



SCHEMATIC DIAGRAM

NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerances are ±.005 (.13 mm) for three place decimals and ±.03 (.8 mm) for two place decimals.
3. Metric equivalents are given for general information only.
4. Unit mounts in panels from .031 (.79 mm) to .19 (4.8 mm) thick.
5. A .174 (4.42 mm) thick spacer is supplied with each switch unit for MIL-P-7788 panel requirements. Spacer places switch mounting flange flush with top of overlay panel. Spacer to be used only for edge-light panels, discard otherwise.
6. Mounting sleeve shall not exceed .760 (19.30 mm) square and shall be reversible to accommodate edge-light panel requirements.
7. Pushbutton designed to prevent incorrect insertion into switch housing.
8. Exact shape of switch optional provided dimensions specified are not exceeded.
9. Available area for legend illumination is .53 square for display type W and .48 square for display type S.

FIGURE 1. Design and construction - Continued.

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REQUIREMENTS:

Design and construction. See figure 1. Basic switches shall be MIL-S-8805/4 (MS24547-1 or -6) or MIL-S-8805/109 (M8805/109-01 or -03) listed in QPL-8805.

Enclosure design: 1 (unsealed).

Temperature characteristic: 1 (-55°C to +85°C).

Vibration grade: 2 (10-500 Hz).

Operation A (Momentary action), B (Alternate action), H (Indicator light).

Illuminated color.

Display type W (White background) - See table I.

Display type S (Sunlight readable) - See table II.

TABLE I. Illuminated color - display type W.

Chromaticity coordinates 1/ 2200° kelvin		
Colors	X	Y
Green	.312 .312 .185 .265	.685 .630 .630 .720
Yellow	.530 .544 .571 .584	.442 SL 2/ .402 SL 2/
Red	.680 .690 .715 .725	.300 .310 .265 .275
Blue	.200 .120 .085 .175	.260 .220 .280 .315
White	.484 Ref.	.415 Ref.
X is not less than .350. Y is not greater than .470. $Y - Y_0$ or $Y_0 - Y$ is not numerically greater than 0.03X. (Y_0 is the Y coordinate of the Planckian radiator for a given X).		

1/ The chromaticities of the diffusing lenses or colored filters expressed as X and Y coordinates on the CIE chromaticity diagram shall lie within areas bounded by the coordinates listed for each color, using a source illuminate of 2200°K.

2/ SL = spectrum locus (where intersected by other of coordinate pair).

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TABLE II. Illuminated color - display type S. 1/

Colors	X	Y
Red	.695	.285
	.710	SL 2/
	.665	.325
	.660	SL 2/
Green	.300	SL 2/
	.300	.600
	.380	.600
	.380	SL 2/
Yellow	.562	.415
	.570	SL 2/
	.596	.382
	.605	SL 2/
Blue	.230	.420
	.230	.420
	.320	.350
	.320	.420
White	.400	.420
	.460	.420
	.400	.380
	.460	.380

1/ The chromaticities of the colors are expressed as "X" and "Y" coordinates of the CIE chromaticity diagram and are within the limits bound by the coordinates listed for each color.

2/ SL - spectrum locus (where intersected by other coordinate pair).

Display type:

W (White until illuminated, then background appears in color).

S (Sunlight readable).

Material:

Housing: Corrosion resistant steel or aluminum alloy.

Weight: 1.6 ounces maximum.

Lamps:

Display type W pushbutton shall accommodate four lamps (T-1 3/4 midget flange base) except lens configuration O (see figure 3) shall accommodate only two lamps.

Display type S (sunlight readable) illumination is by four 5-volt, T-1, .15 MSCP lamps. Other T-1 lamps are available, with different illumination results. Lamps shall be acquired separately.

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Operating characteristics:

Actuation force: 2 to 5 pounds.

Actuation travel: .160 ±.031 inch.

Lens extraction force: 1 to 4 pounds.

Shock: 75 G sawtooth (MIL-STD-202, method 213, test condition H).

Moisture resistance. Following step 6 of the final cycle of the moisture resistance test, while switches are still wet, the insulation resistance measurement shall be not less than 5 megohms.

Electrical endurance: See tables III and IV for electrical load conditions.

TABLE III. Electrical ratings - silver contacts.

Load	Sea level		50,000 feet 28 V dc
	28 V dc	115 V ac, 60 Hz	
	(Amperes)	(Amperes)	(Amperes)
Resistive	7	7	4
Inductive	4	7	2.5
Lamp	2.5	2	2.5

TABLE IV. Electrical ratings - gold contacts.

Load	Sea level		50,000 feet	
	Power circuit 28 V dc	Electronic logic 5 V dc	Power circuit 28 V dc	Electronic logic 5 V dc
	(Amperes)	(Amperes)	(Amperes)	(Amperes)
Resistive	1.0	0.01	1.0	0.01
Inductive	0.5	---	0.5	---

Low level life: Applicable to gold contact switches.

EMI/RFI shielding: Shielding efficiency test applicable, except the shielding attenuation shall be not less than 47 dB at 100 MHz, 39 dB at 400 MHz and 33 dB at 1,000 mHz.

Luminance:

Display type W. See table V.

Display type S: Sunlight readability contrast requirements apply.

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TABLE V. Luminance for display type W. 1/

Color	Color code	Brightness (foot-lamberts)	
		Without EMI screen 2/	With EMI screen 3/
Blue	B	8.0	4.0
Green	G	50.0	25.0
Red	R	30.0	15.0
White	W	180.0	90.0
Yellow	Y	300.0	150.0

1/ When illuminated by two MS25237 lamps of .30 ±.05 mean spherical candlepower at 28 volts dc. Readings are for lens configuration 0.
2/ Minimum average.
3/ For information only.

Sunlight readability: Applicable to display type S switches, see table VII. Units shall be tested for $\phi_1 = 45°$ and $\phi_2 = 0°$ only. This procedure does not test readability under specularly reflective (glare) condition. The average contrast ratio of each legend character to the background shall be 0.6 minimum in the lighted legend condition. In the unlighted legend condition, the average contrast ratio of each legend character to the background shall have an absolute value less than or equal to 0.1.

Marking Terminals and basic switch identification shall be marked as shown on figure 1.

Qualification inspection

Group submission: All part numbers will be approved for inclusion in the qualified products list upon successful completion of testing specified in table VI. Table VI is based on use of qualified MIL-S-8805/4 or MIL-S-8805/109 basic switches in the assembly.

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TABLE VI. Qualification inspection - group submission.

Inspection 1/	Sample
Group I	All samples
Group II	M22885/67-490RG 3/ (4 samples)
Group III	M22885/67-114RG (2 samples)
Group VI 2/	M22885/67-114RG (2 samples) M22885/67-442RG 3/ (2 samples)
Group VII 4/	M22885/67-490RG 3/ (2 samples)
Group VIII	Type S display: 2 pushbuttons of each color (10 samples), of lens configuration 0, and with EMI/RFI shielding. 3/ 5/ Type W display: 2 pushbuttons of each color (10 samples), of lens configuration 0, and without EMI/RFI shielding. 5/

1/ Inspection groups in accordance with qualification inspection table of MIL-S-22885.

2/ Electrical endurance:
Silver contacts - test resistive load, ac, only.
Gold contacts - test sea level resistive, 28 V dc only.
Lamps used during electrical endurance test shall be MS25237-387 for type W display and MS24515-718 for type S display.

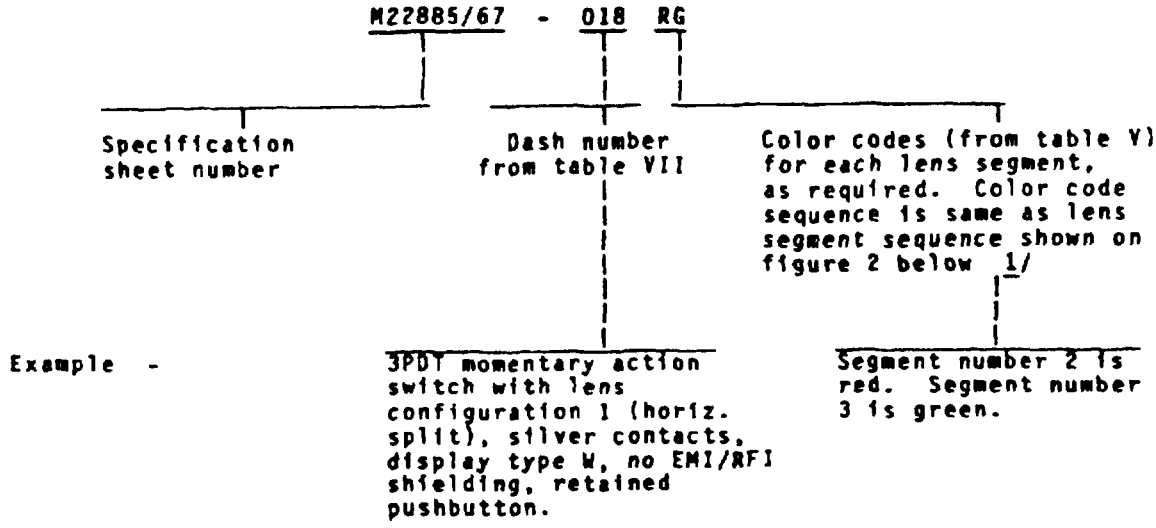
3/ Legend shall be in accordance with MIL-S-22885 sunlight readability test paragraph.

4/ Low level life test not required.

5/ Pushbuttons shall be assembled to any representative switch body.

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Part numbers: Part numbers are assigned as illustrated below. Part numbers do not include legend description. Acquire switches with legends as category II items in accordance with MIL-S-22885.



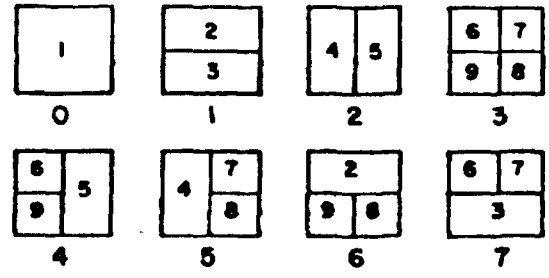


FIGURE 2 Lens configuration and segment numbers.

1/ Part number constructed as shown replace and are interchangeable with previous part numbers which included segment numbers in addition to color codes (example: M22885/67-018-2R3G).

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TABLE VII. Dash numbers.

Dash number	Circuit	Operation	Lens configuration (see figure 2)	Contact material	Display type	EMI/RFI shielding	Retained pushbutton 1/
009	2PDT	Momentary	0	Silver	W	No	Yes
010			1				
011			2				
012			3				
013			4				
014			5				
015			6				
016			7				
017	3PDT	Momentary	0	Silver	W	No	Yes
018			1				
019			2				
020			3				
021			4				
022			5				
023			6				
024			7				
113	4PDT	Momentary	0	Silver	W	No	Yes
114			1				
115			2				
116			3				
117			4				
118			5				
119			6				
120			7				
033	2PDT	Alternate	0	Silver	W	No	Yes
034			1				
035			2				
036			3				
037			4				
038			5				
039			6				
040			7				
041	3PDT	Alternate	0	Silver	W	No	Yes
042			1				
043			2				
044			3				
045			4				
046			5				
047			6				
048			7				
121	4PDT	Alternate	0	Silver	W	No	Yes
122			1				
123			2				
124			3				
125			4				
126			5				
127			6				
128			7				

See footnotes at end of table.

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TABLE VII. Dash numbers - Continued.

Dash number	Circuit	Operation	Lens configuration (see figure 2)	Contact material	Display type	EMI/RFI shielding	Retained pushbutton 1/
129	Lamp circuit only	Light, indicator	0	N/A	W	No	Yes
130			1				
131			2				
132			3				
133			4				
134			5				
135			6				
136			7				
137	2PDT	Momentary	0	Silver	W	Yes	Yes
138			1				
139			2				
140			3				
141			4				
142			5				
143			6				
144			7				
145	3PDT	Momentary	0	Silver	W	Yes	Yes
146			1				
147			2				
148			3				
149			4				
150			5				
151			6				
152			7				
153	4PDT	Momentary	0	Silver	W	Yes	Yes
154			1				
155			2				
156			3				
157			4				
158			5				
159			6				
160			7				
161	2PDT	Alternate	0	Silver	W	Yes	Yes
162			1				
163			2				
164			3				
165			4				
166			5				
167			6				
168			7				
169	3PDT	Alternate	0	Silver	W	Yes	Yes
170			1				
171			2				
172			3				
173			4				
174			5				
175			6				
176			7				

See footnotes at end of table.

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TABLE VII. Dash numbers - Continued.

Dash number	Circuit	Operation	Lens configuration (see figure 2)	Contact material	Display type	EMI/RFI shielding	Retained pushbutton 1/
177			0				
178			1				
179			2				
180	4PDT	Alternate	3	Silver	W	Yes	Yes
181			4				
182			5				
183			6				
184			7				
185			0				
186			1				
187	Lamp		2				
188	circuit	Light,	3	N/A	W	Yes	Yes
189	only	indicator	4				
190			5				
191			6				
192			7				
193			0				
194			1				
195			2				
196	2PDT	Momentary	3	Gold	W	No	Yes
197			4				
198			5				
199			6				
200			7				
201			0				
202			1				
203			2				
204	3PDT	Momentary	3	Gold	W	No	Yes
205			4				
206			5				
207			6				
208			7				
209			0				
210			1				
211			2				
212	4PDT	Momentary	3	Gold	W	No	Yes
213			4				
214			5				
215			6				
216			7				
217			0				
218			1				
219			2				
220	2PDT	Alternate	3	Gold	W	No	Yes
221			4				
222			5				
223			6				
224			7				

See footnotes at end of table.

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TABLE VII. Dash numbers - Continued.

Dash number	Circuit	Operation	Lens configuration (see figure 2)	Contact material	Display type	EMI/RFI shielding	Retained pushbutton 1/
225	3PDT	Alternate	0	Gold	W	No	Yes
226			1				
226			2				
228			3				
229			4				
230			5				
231			6				
232			7				
233	4PDT	Alternate	0	Gold	W	No	Yes
234			1				
235			2				
236			3				
237			4				
238			5				
239			6				
240			7				
241	2PDT	Momentary	0	Gold	W	Yes	Yes
242			1				
243			2				
244			3				
245			4				
246			5				
247			6				
248			7				
249	3PDT	Momentary	0	Gold	W	Yes	Yes
250			1				
251			2				
252			3				
253			4				
254			5				
255			6				
256			7				
257	4PDT	Momentary	0	Gold	W	Yes	Yes
258			1				
259			2				
260			3				
261			4				
262			5				
263			6				
264			7				
265	2PDT	Alternate	0	Gold	W	Yes	Yes
266			1				
267			2				
268			3				
269			4				
270			5				
271			6				
272			7				

See footnotes at end of table.

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TABLE VII. Dash numbers - Continued.

Dash number	Circuit	Operation	Lens configuration (see figure 2)	Contact material	Display type	EMI/RFI shielding	Retained pushbutton 1/
273	3PDT	Alternate	0	Gold	W	Yes	Yes
274			1				
275			2				
276			3				
277			4				
278			5				
279			6				
280			7				
281	4PDT	Alternate	0	Gold	W	Yes	Yes
282			1				
283			2				
284			3				
285			4				
286			5				
287			6				
288			7				
289	2PDT	Momentary	0	Silver	S	No	Yes
290			1				
291			2				
292			3				
293			4				
294			5				
295			6				
296			7				
297	3PDT	Momentary	0	Silver	S	No	Yes
298			1				
299			2				
300			3				
301			4				
302			5				
303			6				
304			7				
305	4PDT	Momentary	0	Silver	S	No	Yes
306			1				
307			2				
308			3				
309			4				
310			5				
311			6				
312			7				
313	2PDT	Alternate	0	Silver	S	No	Yes
314			1				
315			2				
316			3				
317			4				
318			5				
319			6				
320			7				

See footnotes at end of table.

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TABLE VII Dash numbers - Continued.

Dash number	Circuit	Operation	Lens configuration (see figure 2)	Contact material	Display type	EMI/RFI shielding	Retained pushbutton 1/
321 322 323 324 325 326 327 328	3PDT	Alternate	0 1 2 3 4 5 6 7	Silver	S	No	Yes
329 330 331 332 333 334 335 336	4PDT	Alternate	0 1 2 3 4 5 6 7	Silver	S	No	Yes
337 338 339 340 341 342 343 344	Lamp circuit only	Light, indicator	0 1 2 3 4 5 6 7	N/A	S	No	Yes
345 346 347 348 349 350 351 352	2PDT	Momentary	0 1 2 3 4 5 6 7	Silver	S	Yes	Yes
353 354 355 356 357 358 359 360	3PDT	Momentary	0 1 2 3 4 5 6 7	Silver	S	Yes	Yes
361 362 363 364 365 366 367 368	4PDT	Momentary	0 1 2 3 4 5 6 7	Silver	S	Yes	Yes

See footnotes at end of table.

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TABLE VII. Dash numbers - Continued.

Dash number	Circuit	Operation	Lens configuration (see figure 2)	Contact material	Display type	EMI/RFI shielding	Retained pushbutton 1/
369	2PDT	Alternate	0	Silver	S	Yes	Yes
370			1				
371			2				
372			3				
373			4				
374			5				
375			6				
376			7				
377	3PDT	Alternate	0	Silver	S	Yes	Yes
378			1				
379			2				
380			3				
381			4				
382			5				
383			6				
384			7				
385	4PDT	Alternate	0	Silver	S	Yes	Yes
386			1				
387			2				
388			3				
389			4				
390			5				
391			6				
392			7				
393	Lamp circuit only	Light, indicator	0	N/A	S	Yes	Yes
394			1				
395			2				
396			3				
397			4				
398			5				
399			6				
400			7				
401	2PDT	Momentary	0	Gold	S	No	Yes
402			1				
403			2				
404			3				
405			4				
406			5				
407			6				
408			7				
409	3PDT	Momentary	0	Gold	S	No	Yes
410			1				
411			2				
412			3				
413			4				
414			5				
415			6				
416			7				

See footnotes at end of table.

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TABLE VII. Dash numbers - Continued.

Dash number	Circuit	Operation	Lens configuration (see figure 2)	Contact material	Display type	EMI/RFI shielding	Retained pushbutton 1/
417	4PDT	Momentary	0	Gold	S	No	Yes
418			1				
419			2				
420			3				
421			4				
422			5				
423			6				
424			7				
425	2PDT	Alternate	0	Gold	S	No	Yes
426			1				
427			2				
428			3				
429			4				
430			5				
431			6				
432			7				
433	3PDT	Alternate	0	Gold	S	No	Yes
434			1				
435			2				
436			3				
437			4				
438			5				
439			6				
440			7				
441	4PDT	Alternate	0	Gold	S	No	Yes
442			1				
443			2				
444			3				
445			4				
446			5				
447			6				
448			7				
449	2PDT	Momentary	0	Gold	S	Yes	Yes
450			1				
451			2				
452			3				
453			4				
454			5				
455			6				
456			7				
457	3PDT	Momentary	0	Gold	S	Yes	Yes
458			1				
459			2				
460			3				
461			4				
462			5				
463			6				
464			7				

See footnotes at end of table.

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TABLE VII. Dash numbers - Continued.

Dash number	Circuit	Operation	Lens configuration (see figure 2)	Contact material	Display type	EMI/RFI shielding	Retained pushbutton 1/
465	4PDT	Momentary	0	Gold	S	Yes	Yes
466			1				
467			2				
468			3				
469			4				
470			5				
471			6				
472			7				
473	2PDT	Alternate	0	Gold	S	Yes	Yes
474			1				
475			2				
476			3				
477			4				
478			5				
479			6				
480			7				
481	3PDT	Alternate	0	Gold	S	Yes	Yes
482			1				
483			2				
484			3				
485			4				
486			5				
487			6				
488			7				
489	4PDT	Alternate	0	Gold	S	Yes	Yes
490			1				
491			2				
492			3				
493			4				
494			5				
495			6				
496			7				
2/ 001	1PDT	Momentary	0	Silver	W	No	Yes
002			1				
003			2				
004			3				
005			4				
006			5				
007			6				
008			7				
2/ 025	1PDT	Alternate	0	Silver	W	No	Yes
026			1				
027			2				
028			3				
029			4				
030			5				
031			6				
032			7				

See footnotes at end of table.

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TABLE VII. Dash numbers - Continued.

Dash number	Circuit	Operation	Lens configuration (see figure 2)	Contact material	Display type	EMI/RFI shielding	Retained pushbutton 1/
2/ 049	2PDT	Momentary	0	Silver	W	No	No
050			1				
051			2				
052			3				
053			4				
054			5				
055			6				
056			7				
2/ 057	3PDT	Momentary	0	Silver	W	No	No
058			1				
059			2				
060			3				
061			4				
062			5				
063			6				
064			7				
2/ 065	2PDT	Alternate	0	Silver	W	No	No
066			1				
067			2				
068			3				
069			4				
070			5				
071			6				
072			7				
2/ 073	3PDT	Alternate	0	Silver	W	No	No
074			1				
075			2				
076			3				
077			4				
078			5				
079			6				
080			7				
2/ 081	2PDT	Momentary	0	Silver	W	Yes	No
082			1				
083			2				
084			3				
085			4				
086			5				
087			6				
088			7				
2/ 089	3PDT	Momentary	0	Silver	W	Yes	No
090			1				
091			2				
092			3				
093			4				
094			5				
095			6				
096			7				

See footnotes at end of table.

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TABLE VII. Dash numbers - Continued.

Dash number	Circuit	Operation	Lens configuration (see figure 2)	Contact material	Display type	EMI/RFI shielding	Retained pushbutton 1/
2/ 097	2PDT	Alternate	0	Silver	W	Yes	No
098			1				
099			2				
100			3				
101			4				
102			5				
103			6				
104			7				
2/ 105	3PDT	Alternate	0	Silver	W	Yes	No
106			1				
107			2				
108			3				
109			4				
110			5				
111			6				
112			7				

1/ Retained pushbutton shall be capable of having lamps replaced but is restricted in complete removal from switch housing.

2/ Inactive for new design.

NOTE: 800 and 900 series dash numbers, for switch housing only and pushbutton only, have been canceled without replacement.

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NOTE: Revision letter are not use to denote changes due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AV, MI
Navy - AS, OS
Air Force - 99
DLA - ES

User activities:
Army - AR
Navy - MC, SH
Air Force - 11, 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5930-1365-01)

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